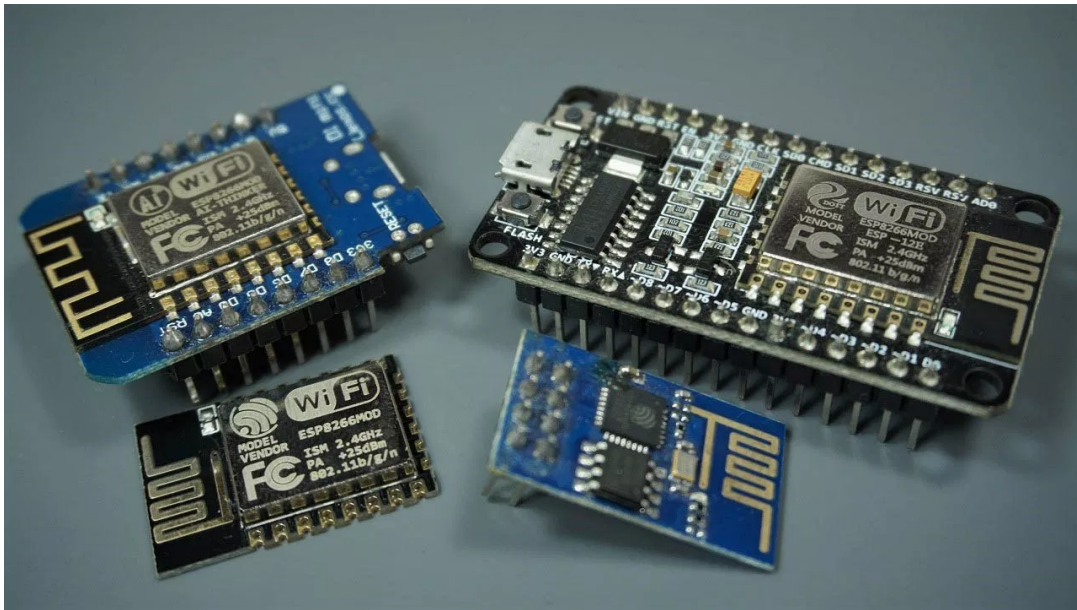


ESP8266 Pinout Reference: Which GPIO pins should you use?

This article is a guide for the ESP8266 GPIOs: pinout diagrams, their functions and how to use them.



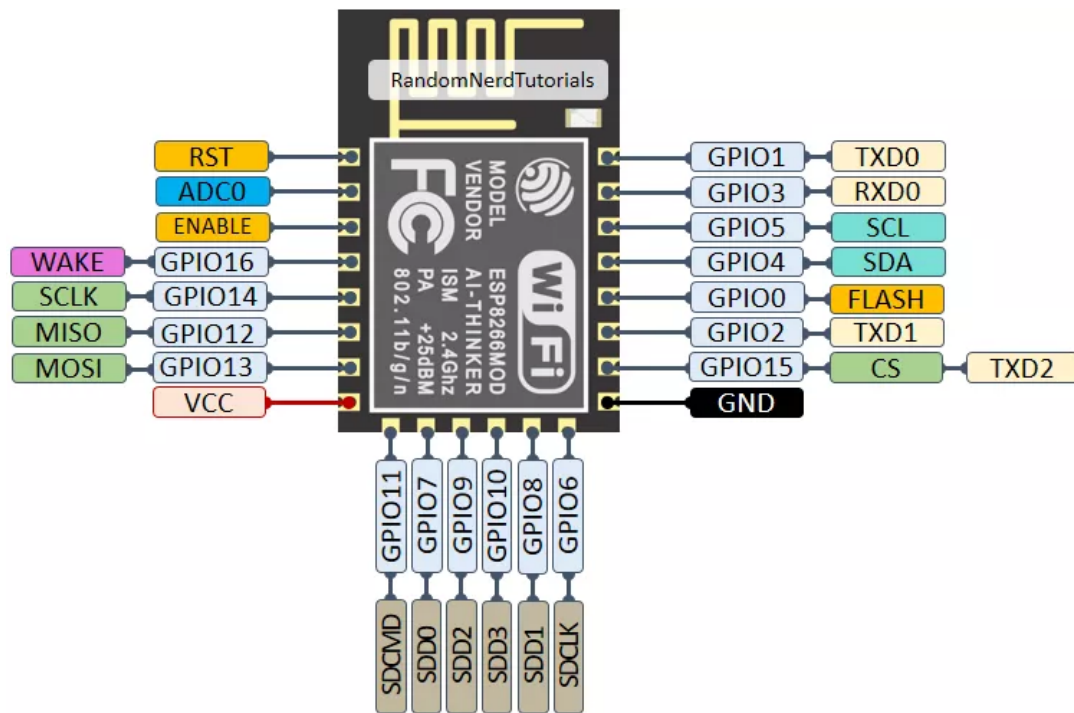
The ESP8266 12-E chip comes with 17 GPIO pins. Not all GPIOs are exposed in all ESP8266 development boards, some GPIOs are not recommended to use, and others have very specific functions.

With this guide, you'll learn how to properly use the ESP8266 GPIOs and avoid hours of frustration by using the most suitable pins for your projects.

We also have a guide for the ESP32 GPIOs: [ESP32 Pinout Reference: Which GPIO pins should you use?](#)

ESP8266 12-E Chip Pinout

The following figure illustrates the ESP8266 12-E chip pinout. Use this diagram if you're using an ESP8266 bare chip in your projects.



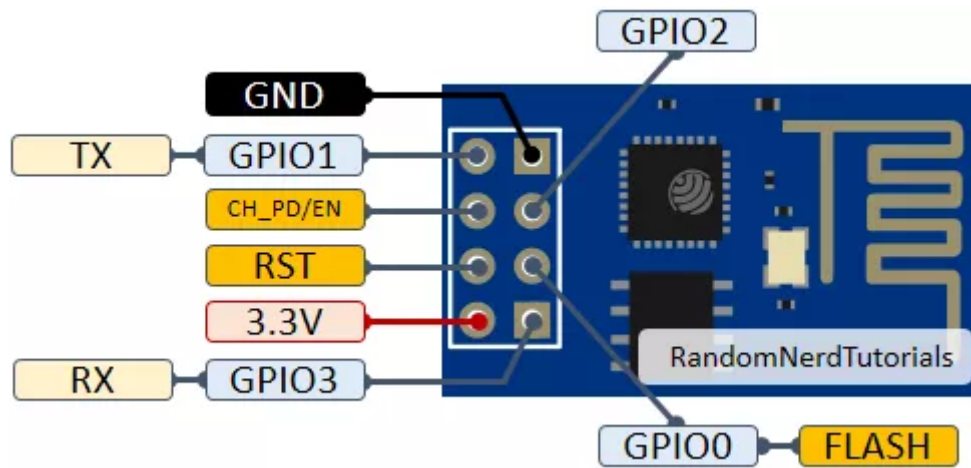
Note: not all GPIOs are accessible in all development boards, but each specific GPIO works in the same way regardless of the development board you're using. If you're just getting started with the ESP8266, we recommend reading our guide: [Getting Started with the ESP8266](#).

At the moment, there are a wide variety of development boards with the ESP8266 chip that differ in the number of accessible GPIOs, size, form factor, etc...

The most widely used ESP8266 boards are the [ESP-01](#), [ESP8266-12E NodeMCU Kit](#), and the [Wemos D1 Mini](#). For a comparison of these board, you can read this guide: [ESP8266 Wi-Fi Development Boards comparison](#).

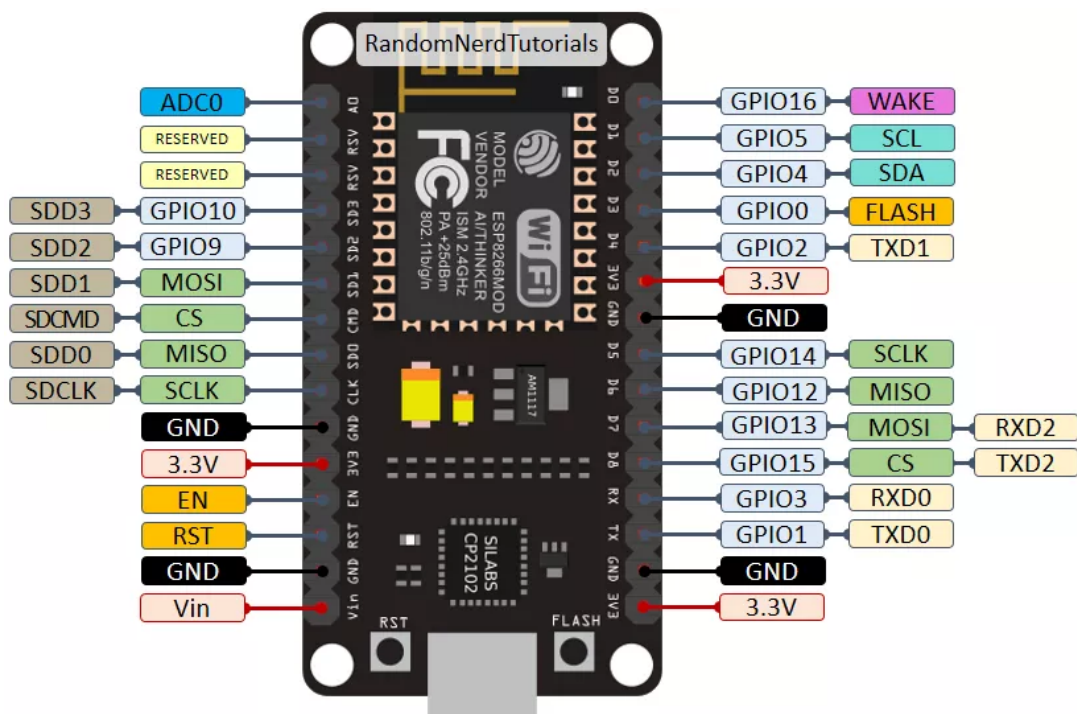
ESP8266-01 Pinout

If you're using an ESP8266-01 board, you can use the following GPIO diagram as a reference.



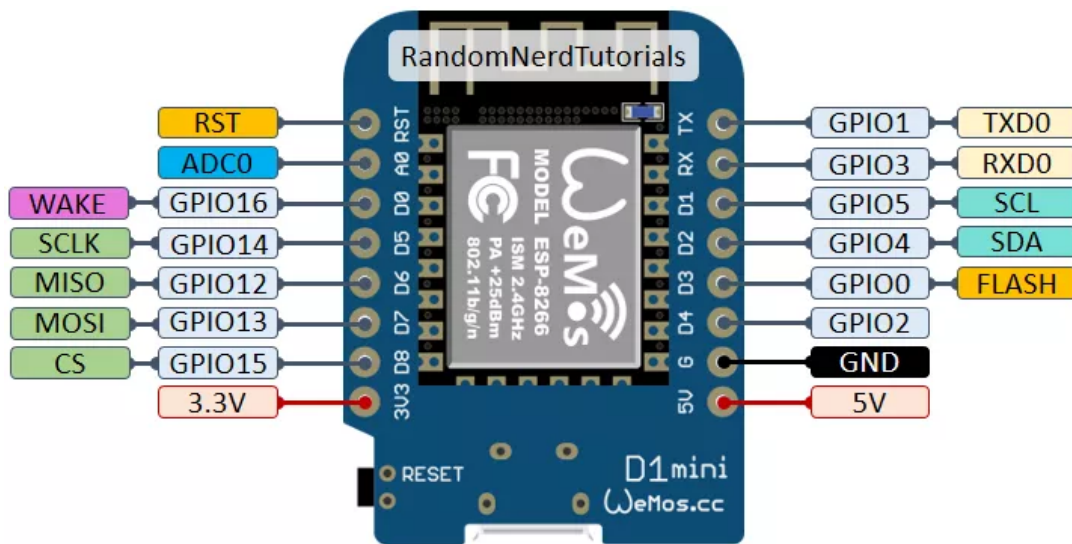
ESP8266 12-E NodeMCU Kit

The ESP8266 12-E NodeMCU kit pinout diagram is shown below.



Wemos D1 Mini Pinout

The following figure shows the WeMos D1 Mini pinout.



Download PDF with ESP8266 Pinout Diagrams

We've put together a handy PDF that you can download and print, so you always have the ESP8266 diagrams next to you:

[Download PDF Pinout Diagrams »](#)

ESP8266 Peripherals

The ESP8266 peripherals include:

- 17 GPIOs
- SPI
- I2C (implemented on software)
- I2S interfaces with DMA
- UART
- 10-bit ADC

Best Pins to Use – ESP8266

One important thing to notice about ESP8266 is that the GPIO number doesn't match the label on the board silkscreen. For example, D0 corresponds to GPIO16 and D1 corresponds to GPIO5.

The following table shows the correspondence between the labels on the silkscreen and the GPIO number as well as what pins are the best to use in your projects, and which ones you need to be cautious.

The pins highlighted in green are OK to use. The ones highlighted in yellow are OK to use, but you need to pay attention because they may have unexpected behavior mainly at boot. The pins highlighted in red are not recommended to use as inputs or outputs.

Label	GPIO	Input	Output	Notes
D0	GPIO16	no interrupt	no PWM or I2C support	HIGH at boot used to wake up from deep sleep
D1	GPIO5	OK	OK	often used as SCL (I2C)
D2	GPIO4	OK	OK	often used as SDA (I2C)
D3	GPIO0	pulled up	OK	connected to FLASH button, boot fails if pulled LOW
D4	GPIO2	pulled up	OK	HIGH at boot connected to on-board LED, boot fails if pulled LOW
D5	GPIO14	OK	OK	SPI (SCLK)
D6	GPIO12	OK	OK	SPI (MISO)
D7	GPIO13	OK	OK	SPI (MOSI)
D8	GPIO15	pulled to GND	OK	SPI (CS) Boot fails if pulled HIGH
RX	GPIO3	OK	RX pin	HIGH at boot
TX	GPIO1	TX pin	OK	HIGH at boot debug output at boot, boot fails if pulled LOW
A0	ADC0	Analog Input	X	

Continue reading for a more detailed and in-depth analysis of the ESP8266 GPIOs and its functions.

GPIOs connected to the Flash Chip

GPIO6 to GPIO11 are usually connected to the flash chip in ESP8266 boards. So, these pins are not recommended to use.

Pins used during Boot

The ESP8266 can be prevented from booting if some pins are pulled LOW or HIGH. The following list shows the state of the following pins on BOOT:

- **GPIO16:** pin is high at BOOT
- **GPIO0:** boot failure if pulled LOW
- **GPIO2:** pin is high on BOOT, boot failure if pulled LOW
- **GPIO15:** boot failure if pulled HIGH
- **GPIO3:** pin is high at BOOT
- **GPIO1:** pin is high at BOOT, boot failure if pulled LOW
- **GPIO10:** pin is high at BOOT
- **GPIO9:** pin is high at BOOT

Pins HIGH at Boot

There are certain pins that output a 3.3V signal when the ESP8266 boots. This may be problematic if you have relays or other peripherals connected to those GPIOs. The following GPIOs output a HIGH signal on boot:

- GPIO16
- GPIO3
- GPIO1
- GPIO10
- GPIO9

Additionally, the other GPIOs, except GPIO5 and GPIO4, can output a low-voltage signal at boot, which can be problematic if these are connected to transistors or relays. You can [read this article](#) that investigates the state and behavior of each GPIO on boot.

GPIO4 and GPIO5 are the most safe to use GPIOs if you want to operate relays.

Analog Input

The ESP8266 only supports analog reading in one GPIO. That GPIO is called **ADC0** and it is usually marked on the silkscreen as **A0**.

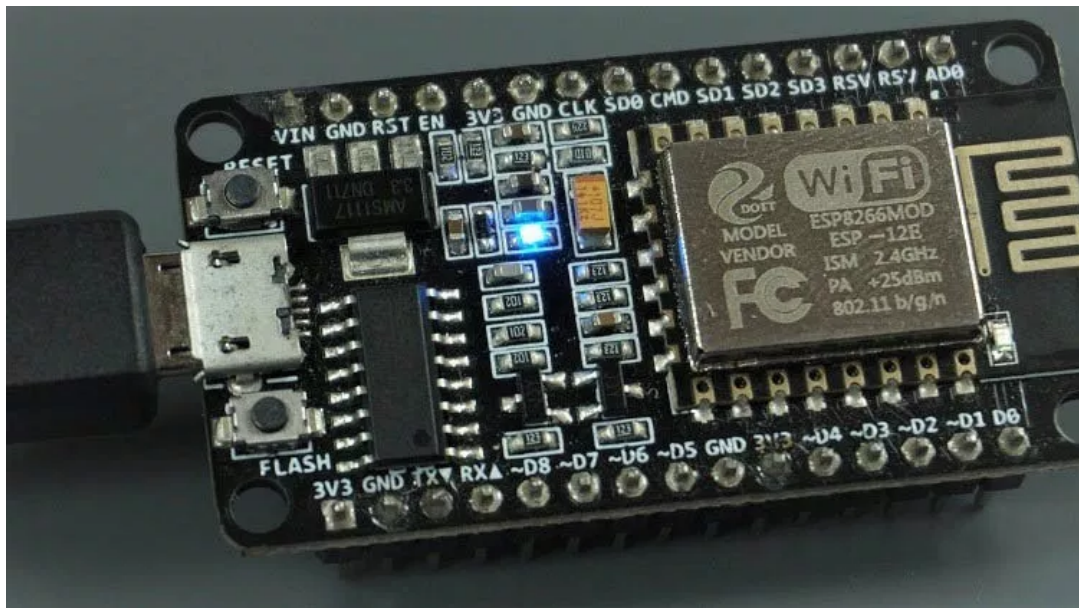
The maximum input voltage of the ADC0 pin is 0 to 1V if you're using the ESP8266 bare chip. If you're using a development board like the ESP8266 12-E NodeMCU kit, the voltage input range is 0 to 3.3V because these boards contain an internal voltage divider.

You can learn how to use analog reading with the ESP8266 with the following guide:

- [ESP8266 ADC – Read Analog Values with Arduino IDE, MicroPython and Lua](#)

On-board LED

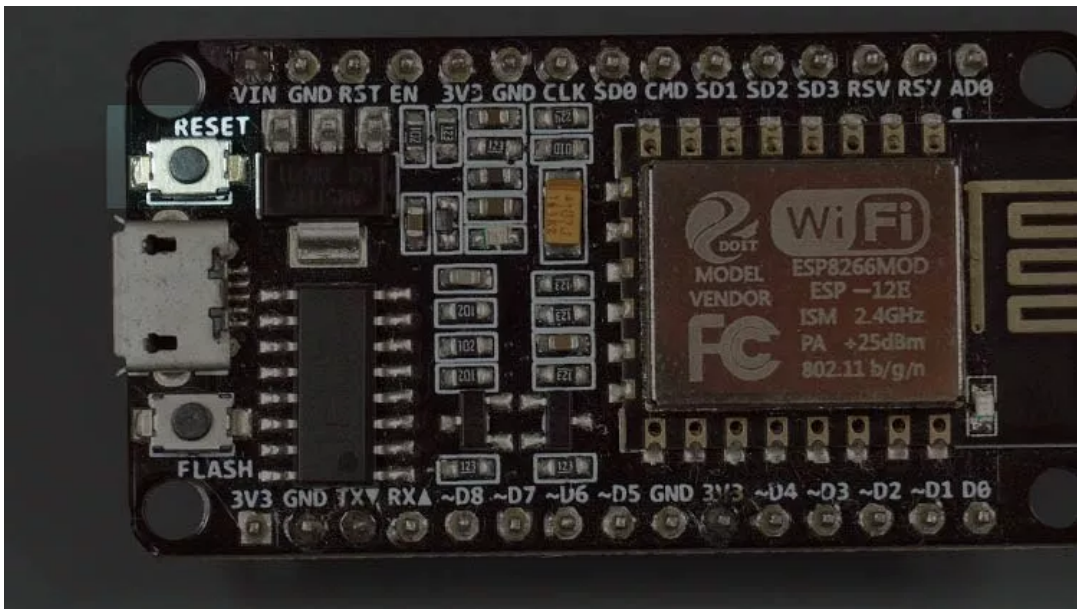
Most of the ESP8266 development boards have a built-in LED. This LED is usually connected to GPIO2.



The LED is connected to a pull-down resistor, so when you send a HIGH signal the LED turns off.

RST Pin

When the RST pin is pulled LOW, the ESP8266 resets. This is the same as pressing the on-board RESET button.



GPIO0

When GPIO0 is pulled LOW, it sets the ESP8266 into bootloader mode. This is the same as pressing the on-board FLASH/BOOT button.



GPIO16

GPIO16 can be used to wake up the ESP8266 from deep sleep. To wake up the ESP8266 from deep sleep, GPIO16 should be connected to the RST pin. Learn how to put the ESP8266 into deep sleep mode:

- [ESP8266 Deep Sleep with Arduino IDE](#)
- [ESP8266 Deep Sleep with MicroPython](#)

I2C

The ESP8266 doesn't have hardware I2C pins, but it can be implemented in software. So you can use any GPIOs as I2C. Usually, the following GPIOs are used as I2C pins:

- **GPIO5:** SCL
- **GPIO4:** SDA

SPI

The pins used as SPI in the ESP8266 are:

- **GPIO12:** MOSI
- **GPIO13:** MISO
- **GPIO14:** SCLK
- **GPIO15:** CS

PWM Pins

ESP8266 allows software PWM in all I/O pins: GPIO0 to GPIO16. PWM signals on ESP8266 have 10-bit resolution. Learn how to use ESP8266 PWM pins:

- [ESP8266 PWM with Arduino IDE](#)
- [ESP8266 PWM with MicroPython](#)

Interrupt Pins

The ESP8266 supports interrupts in any GPIO, except GPIO16.

- [ESP8266 Interrupts and Timers with Arduino IDE](#)

Wrapping Up

We hope you've found this guide for the ESP8266 GPIOs useful. If you have some tips on how to use the ESP8266 GPIOs properly, you can write a comment below.

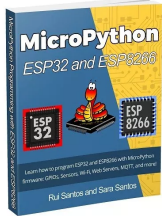
We also have a [similar guide for the ESP32 GPIOs that you can read](#).

If you're getting started with the ESP8266, we have some great content you might be interested in:

- [Home Automation using ESP8266 \(course\)](#)

- [Getting Started with the ESP8266](#)
- [30+ ESP8266 Projects and Tutorials](#)
- [ESP8266 Web Server Tutorial](#)
- [ESP32 vs ESP8266 – Pros and Cons](#)

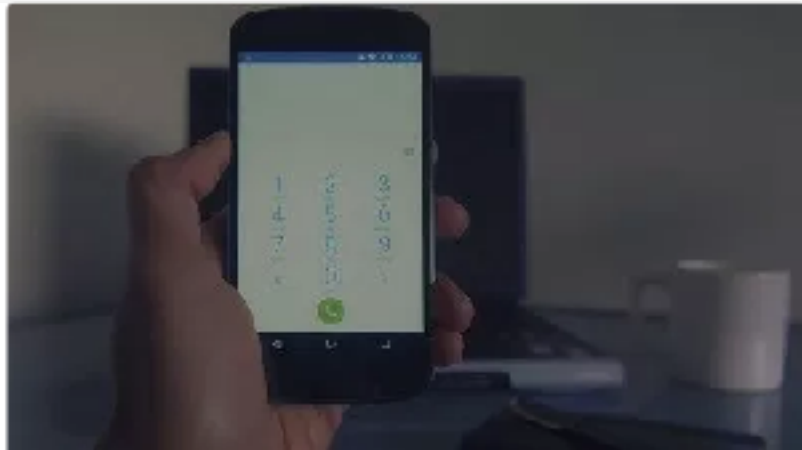
Thanks for reading.



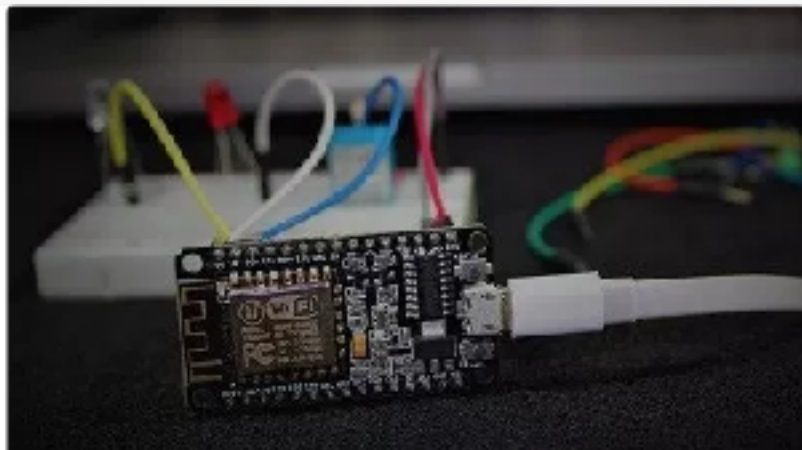
[eBook] MicroPython Programming with ESP32 and ESP8266

Learn how to program and build projects with the ESP32 and ESP8266 using MicroPython firmware [DOWNLOAD »](#)

Recommended Resources

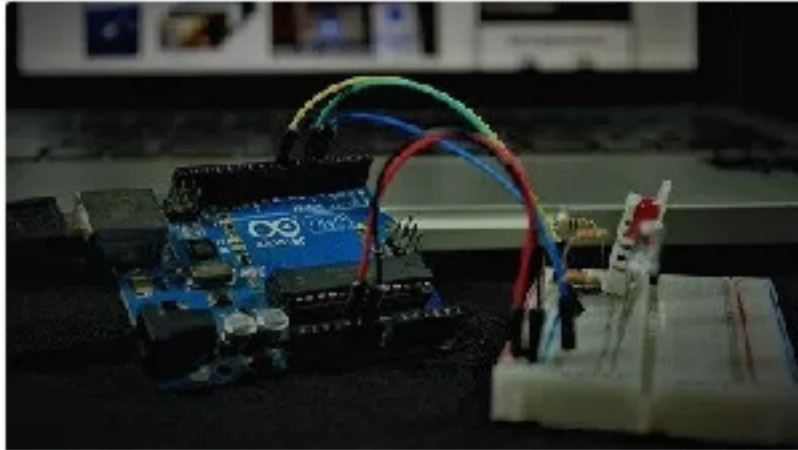


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[Home Automation using ESP8266 eBook and video course »](#) Build IoT

and home automation projects.



[Arduino Step-by-Step Projects](#) » Build 25 Arduino projects with our course, even with no prior experience!

What to Read Next...

[ESP32 with LoRa using Arduino IDE – Getting Started](#)

[ESP8266 ADC – Read Analog Values with Arduino IDE, MicroPython and Lua](#)

[ESP8266 Wi-Fi Button – DIY Amazon Dash Button Clone](#)

[Getting Date and Time with ESP32 on Arduino IDE \(NTP Client\)](#)

[ESP32-CAM Troubleshooting Guide: Most Common Problems Fixed](#)

[ESP8266 DHT11/DHT22 Temperature and Humidity Web Server with Arduino IDE](#)

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38 thoughts on “ESP8266 Pinout Reference: Which GPIO pins should you use?”



Brian

May 6, 2019 at 3:53 pm | Reply

Excellent all in one place info. Thanks



Sara Santos

May 6, 2019 at 5:48 pm | Reply

Thanks 😊



Andrew Wilson

August 25, 2019 at 2:12 pm | Reply

A really comprehensive guide. I've done a lot of trial and error with these devices. I now have all of the insight to design my projects and possibly a commercial product correctly from scratch. I am much indebted.



GurraE

May 6, 2019 at 3:55 pm | Reply

Rui, your site is one of the best there is for learning ESP8266 (& ESP32). I've struggled a lot with projects that won't boot etc. and now I understand why !!!

Your articles are always very professional, very interesting and easy to follow.

I have found many solutions to problems by studying your articles & projects !!

Thank you !!



Sara Santos

May 7, 2019 at 9:28 am | Reply

Hi.

Thank you so much for your nice works.

I'm very happy to see that our readers really enjoy our work. That keeps me motivated to write even better content.

Regards,

Sara



Michele

May 6, 2019 at 4:00 pm | Reply

Great article, thanks !!

It's always convenient to have tables showing the pin outs of these fantastic cards.

I've been trying to flash an ESP-07S with nodemcu firmware lately, but I didn't succeed. Do you have a chart for this card?

Greetings. Mike.



Mike.

May 6, 2019 at 4:12 pm | Reply

Thank you for producing comprehensive guides to ESP8266 devices /boards (also ESP12E), they provide very useful and accessible information sources, without the need for internet searches.



Sara Santos

May 7, 2019 at 9:21 am | Reply

Thank you 😊



Lance Benson

May 6, 2019 at 4:41 pm | Reply

Thanks for this useful guide to the pinouts.

Does the Wemos D1 mini have an onboard voltage divider for the analog input? I ask because I just tried an ACS712 Current Sensor Module, 5V powered, and divided down to the 1V range with a 4K7 & 1K2 resistor divider.

The ACS712 gives a “neutral” reading of half the range, so I was expecting around 512 with nothing connected. Instead I get 170, which would be expected if the A0 input was divided onboard by 3.

(Now what do I need to do to turn a divide by 3 into a divide by 5?)



Dave

May 6, 2019 at 4:50 pm | Reply

This clears up some questions, thanks!

Is there a key or guide for the color tabs used in the pinout diagrams? I understand black and red, but light blue, gray, etc?

Thanks



Sara Santos

May 7, 2019 at 9:20 am | Reply

Hi Dave.

Those are just random colors we've chosen to differentiate the pins.

Pins with similar functions are highlighted with the same color.

For example, SPI is green, I2C is light blue, etc.

Regards,

Sara



Transmic

May 6, 2019 at 6:43 pm | Reply

Wonderful, clear and concise article.

I bought one of your guide and it's written this way.

Pleasant to read and to use!



Sara Santos

May 7, 2019 at 9:13 am | Reply

Thank you so much! 😊

**Geraldo Cartolano**

May 6, 2019 at 6:51 pm | Reply

O trabalho de vcs é perfeito, sempre fico impressionado com a qualidade e precisão de suas postagens.

**Sara Santos**

May 7, 2019 at 9:12 am | Reply

Obrigada 😊

**Nils Skansar**

May 6, 2019 at 7:04 pm | Reply

You are correct, the A0 input is divided. See drawing.
escapequotes.net/esp8266-wemos-d1-mini-pins-and-diagram/
In the middle of the schematic far right you can the the divider.

**Nils Skansar**

May 6, 2019 at 7:23 pm | Reply

Forgot to mention that you can modify the voltage divider on the ACS712 to a 5 to 3 V divider. Here some math comes in. since you will load 1 voltage divider by another. Some testing needed. Final calibration can be done in software.

**Roberto Pangallo**

May 6, 2019 at 8:50 pm | Reply

Great tutorial! Thank you, Sara and Rui.

Roberto

Austria

**Sara Santos**

May 7, 2019 at 9:17 am | Reply

Thank you 😊

**Lance Benson**

May 6, 2019 at 9:13 pm | Reply

Re voltage divider on the D1 Mini, the link below says onboard there are 220K and 100K resistors, so to divide down 5 volts, you add 180K in series to A0.

letscontrolit.com/wiki/index.php/DC_Voltage_divider

**Sara Santos**

May 7, 2019 at 9:17 am | Reply

Thanks for answering.

**denys**

May 6, 2019 at 11:33 pm | Reply

Great resource, thank you.

Your site is one that I recommend to folks often as your courses are so well put together. Please keep it up 😊

**Sara Santos**

May 7, 2019 at 9:16 am | Reply

Thank you so much for recommending our our to your friends 😊

**Logan Byrne**

May 7, 2019 at 12:09 am | Reply

I have an ESP-WROOM-32 board, what pin-out should I follow? I looks kinda like the ESP8266 12-E NodeMCU Kit, pins are differently labelled. Any assistance.

**Sara Santos**

May 7, 2019 at 9:16 am | Reply

Hi Logan.

We have a similar guide for the ESP32. Take a look at it:

<https://randomnerdtutorials.com/esp32-pinout-reference-gpios/>

Regards,

Sara

**Sarawut.c**

May 7, 2019 at 2:28 am | Reply

You've just answer my problem, very good for reference.
Thank you!

**Sara Santos**

May 7, 2019 at 9:15 am | Reply

Thank you.

**Sonny**

May 7, 2019 at 6:18 am | Reply

Excellent guide. Thanks.

**Nuno**

May 8, 2019 at 9:32 am | Reply

Greetings

Excellent guide.

I was struggling with a D1 mini, three switch buttons and four relays.

It seems that I must use an esp32.

Thanks



Sara Santos

May 8, 2019 at 9:50 am | Reply

Yes, it should probably be more appropriate.

We also have a guide about the ESP32 GPIOs that might help you a lot: <https://randomnerdtutorials.com/esp32-pinout-reference-gpios/>

Regards,
Sara



Lance Benson

May 8, 2019 at 12:32 pm | Reply

A D1 mini should be able to handle 3 switch buttons and 4 relays. Note that RX and TX can be made to be GPIOs if need to be. I've been bitten by boot condition needs—specifically, I had a pullup for a DS18B20 on D8, which disabled normal booting.

It's also possible to add an MCP23017 16-bit I2C GPIO expander module (<\$3US on ebay) to gain a lot more I/Os on a D1 Mini (I like the D1 Mini a lot).



Gerard

August 17, 2019 at 6:17 pm | Reply

Hi Sara and Rui,
What is the function of the "EN" pin on NodeMCU?

Cheers,
Gerard



Sara Santos

August 24, 2019 at 10:35 am | Reply

Hi Gerard.

That's the Enable pin for the ESP8266. When it is pulled down, it resets the ESP8266.

Regards,
Sara



Jose Vazquez

August 20, 2019 at 5:32 pm | Reply

Hey guys! This page is very useful!

I'm currently prototyping using a Wemos D1 mini, but the final processor will be an ESP12EX. Since it has more GPIOs available at the bottom (GPIO: 6, 8, 10, 9, 7, 11) do you know if these are usable? I'm planning to create my own breakout board since I don't need the extra stuff NodeMCU has. Thank you!



Sara Santos

August 24, 2019 at 10:18 am | Reply

Hi Jose.

Those pins are usually connected to the ESP8266 flash chip. So, it is better to use other pins.

Regards,
Sara

**Steve**

August 26, 2019 at 11:55 pm | Reply

Thanks so much! Saved me a lot of grief, as I was going to use a few GPIO inputs that would have not worked well.

**Bogdan Coticopol**

October 3, 2019 at 11:13 pm | Reply

The information here is pure gold. Thank you for the detailed explanation on ESP8266!

**Sara Santos**

October 4, 2019 at 9:33 am | Reply

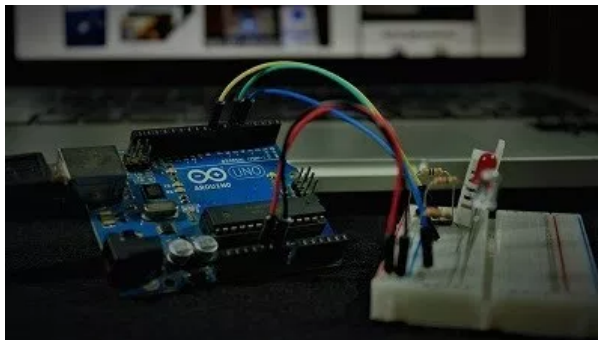
You're welcome 😊

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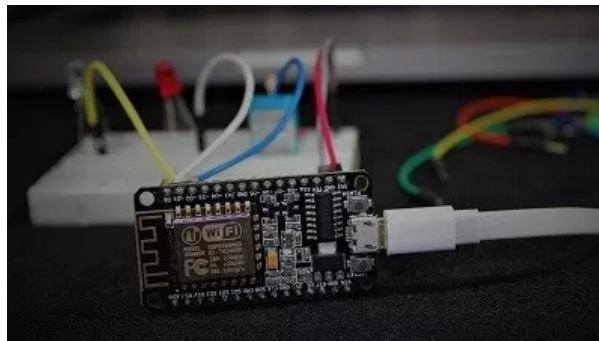


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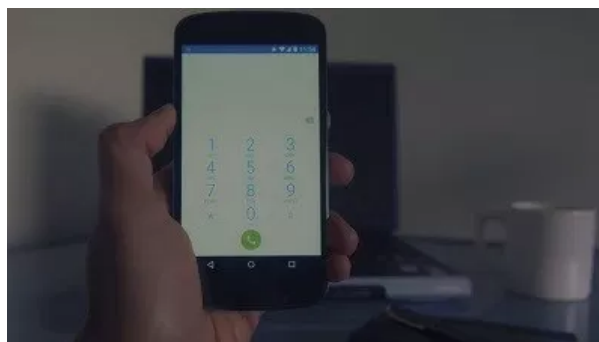
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